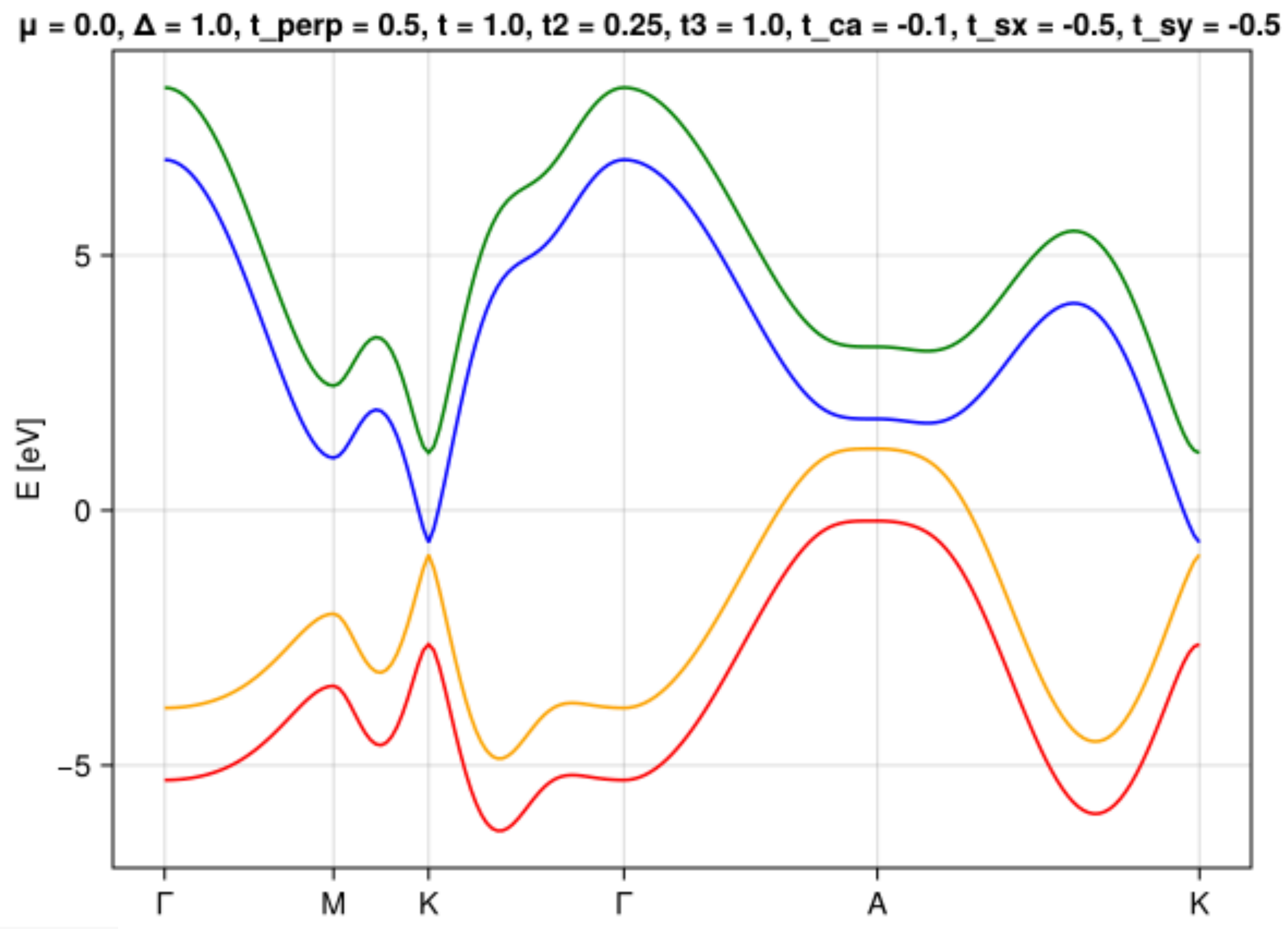
# The spinfull ferroaxial model with KM SOC and spin mixing

$$H_{spinfull} = H_{twobands} \times s_0 + 2t_\lambda u(k) \times \left(\sigma_z \times s_z\right) + t_{sx}(\sigma_0 \times s_x) + t_{sy}(\sigma_0 \times s_y)$$

$$u(k) = \sum_{i=1}^{3} sin(k \cdot \Delta_i) \qquad \Delta_i = [\delta_1 - \delta_2, \delta_2 - \delta_3, \delta_3 - \delta_1]$$




$$H_{2bands} = \begin{pmatrix} -\mu + \Delta + t_2 f_2(k) & t\,f(k) + t_3\,g(k) + t_\perp m(k) \\ h.c. & -\mu - \Delta + t_2 f_2(k) \end{pmatrix}$$

**Same as before**

1st $\quad f(k) = \sum e^{i\vec{k}\cdot\vec{\delta}_i}$

2nd $\quad f_2(k) = e^{i\vec{k}\cdot(\vec{\delta}_1-\vec{\delta}_2)} + e^{i\vec{k}\cdot(\vec{\delta}_2-\vec{\delta}_3)} + e^{i\vec{k}\cdot(\vec{\delta}_3-\vec{\delta}_1)} + e^{i\vec{k}\cdot(-\vec{\delta}_1+\vec{\delta}_2)} + e^{i\vec{k}\cdot(-\vec{\delta}_2+\vec{\delta}_3)} + e^{i\vec{k}\cdot(-\vec{\delta}_3+\vec{\delta}_1)}$

3rd (opposite signs) $\quad g(\vec{k}) = e^{i\vec{k}\cdot(2\vec{\delta}_1-\vec{\delta}_2)} + e^{i\vec{k}\cdot(2\vec{\delta}_2-\vec{\delta}_3)} + e^{i\vec{k}\cdot(2\vec{\delta}_3-\vec{\delta}_1)} - e^{i\vec{k}\cdot(2\vec{\delta}_1-\vec{\delta}_3)} - e^{i\vec{k}\cdot(2\vec{\delta}_2-\vec{\delta}_1)} - e^{i\vec{k}\cdot(2\vec{\delta}_3-\vec{\delta}_2)}$

Perpendicular $\quad m(k) = \sum e^{i\vec{k}\cdot(\vec{\delta}_i+\vec{a}_3)} + \sum e^{i\vec{k}\cdot(\vec{\delta}_i-\vec{a}_3)}$

$$a_1 = a/2 * [3,\sqrt{3},0]$$
$$a2 = a/2 * [3,-\sqrt{3},0]$$
$$a3 = a * [0,0,1]$$
$$a = c = 1\mathring{A}$$

Parameters

$\Delta = 1$ eV

$t = t_3 = 1$ eV

$t_\perp = 0.5$ eV

t_2 = 0.25 eV

$\mu = 0$ eV

$t_\lambda = -0.1$ eV

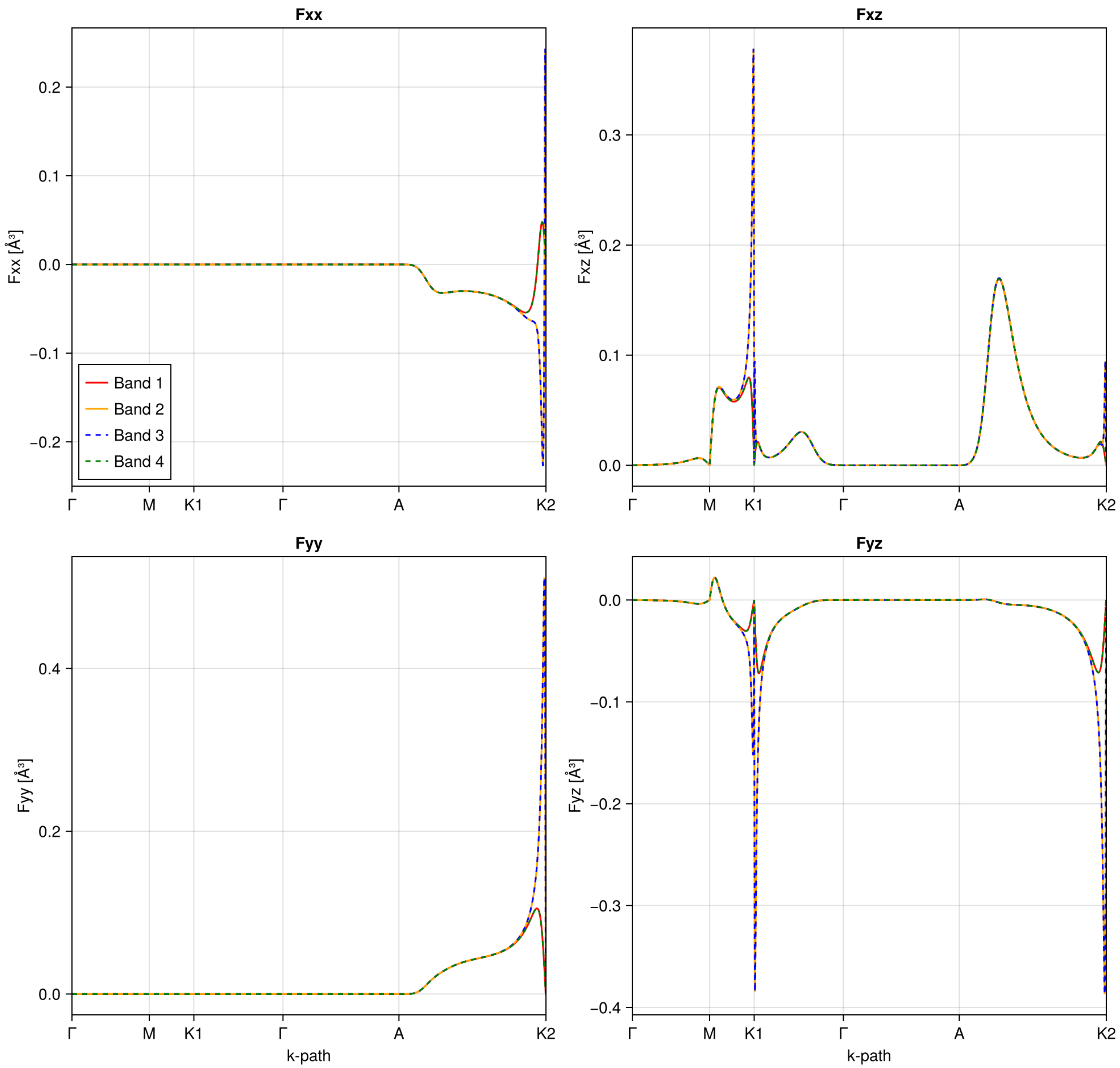
$t_{sx} = -0.5$eV

$t_{sy} = -0.5$eV

# K-resolved calculations: Fij

## QM

Δ = 1.0, t_perp = 0.5, t = 1.0, t2 = 0.25, t3 = 1.0, tλ = -0.1, tsx = -0.5, tsy = -0.5 [eV]

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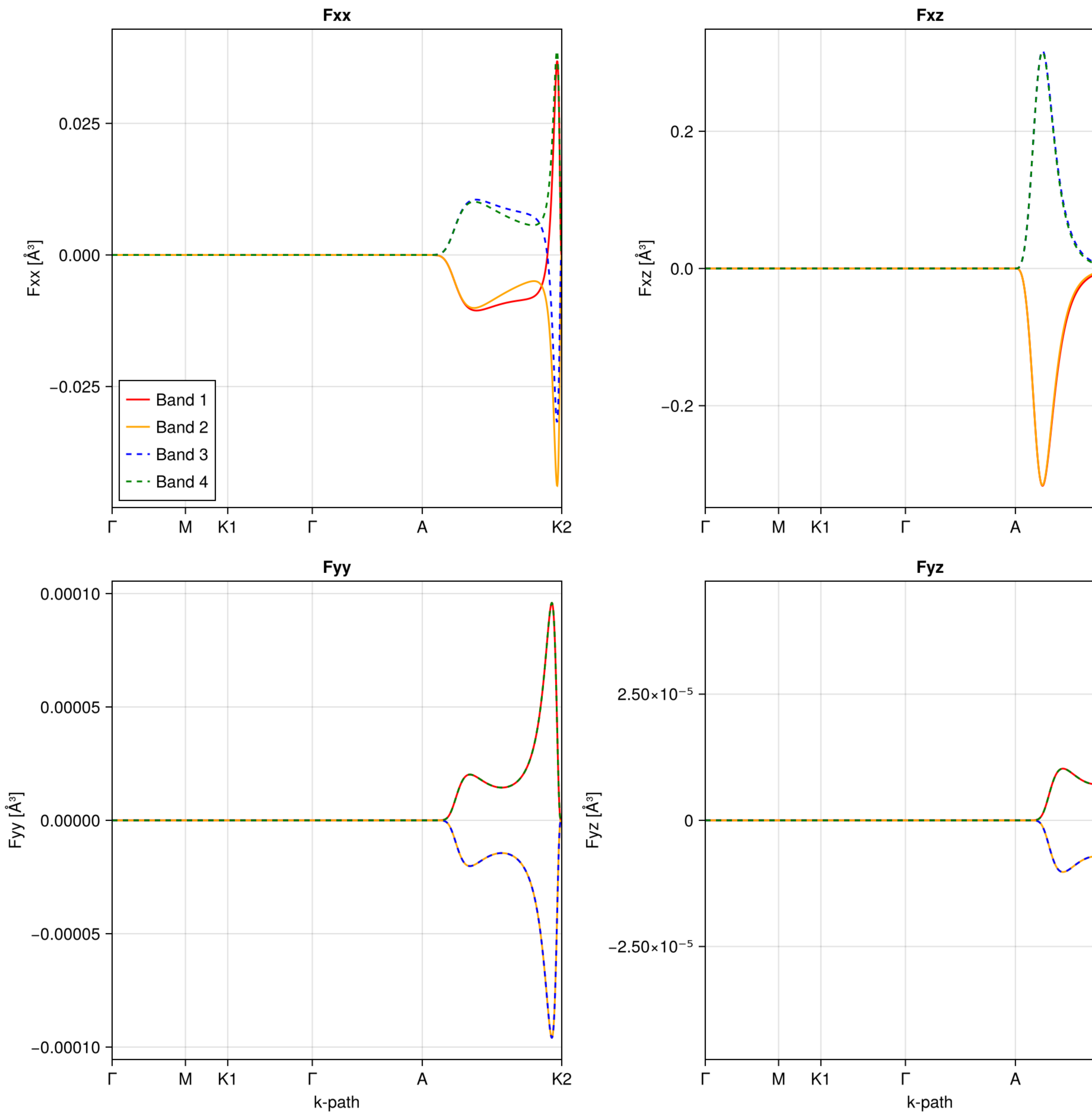

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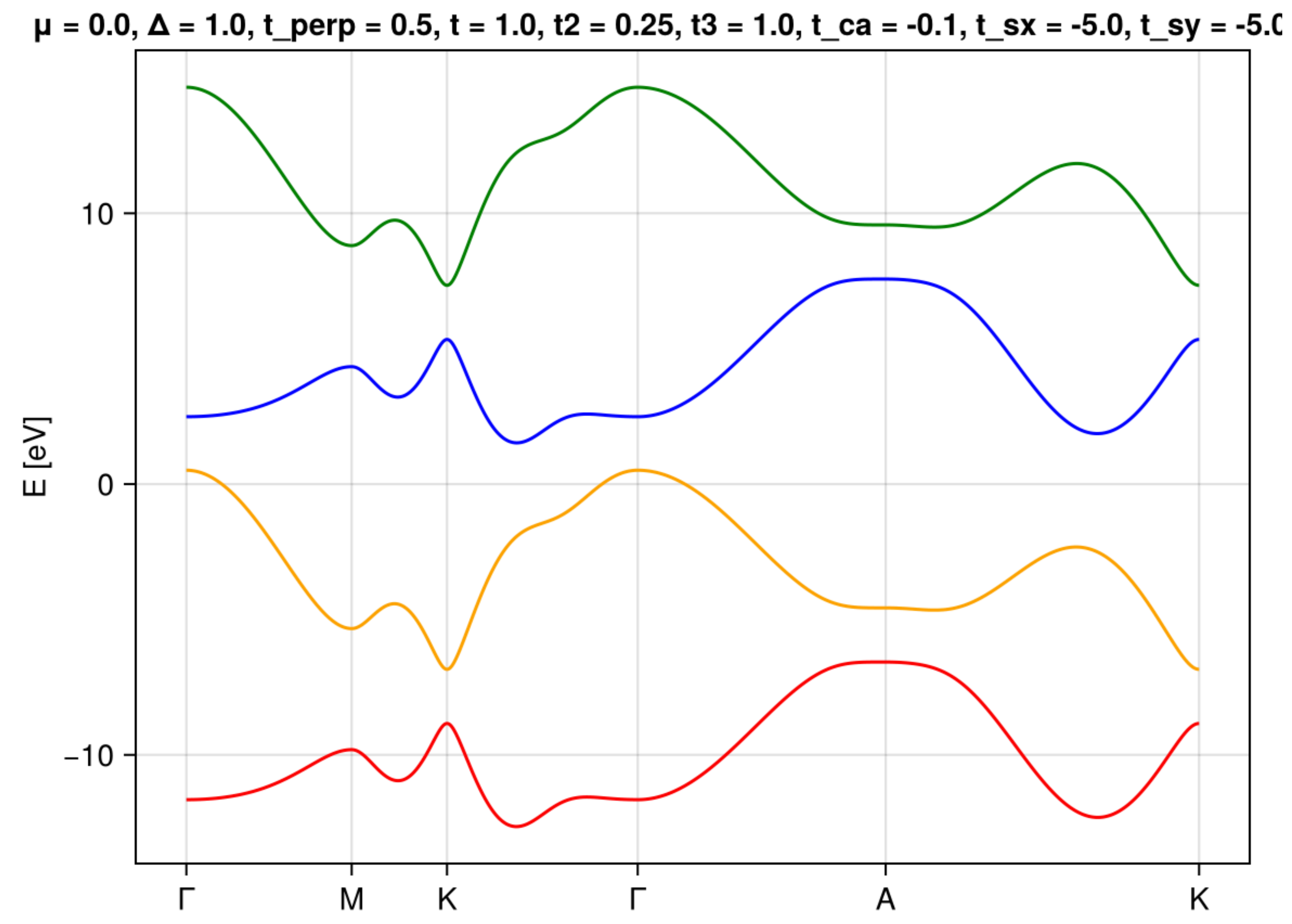


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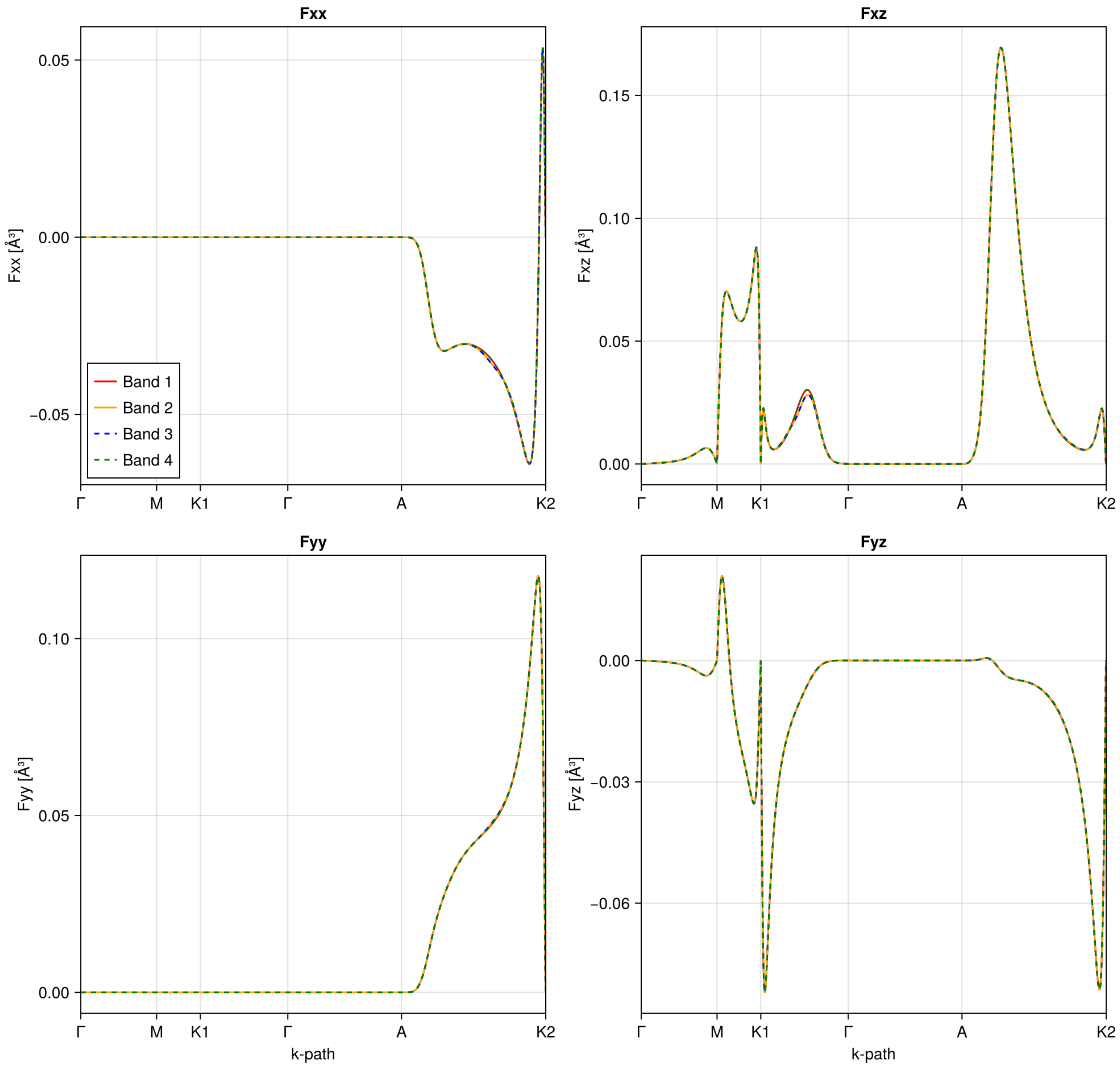


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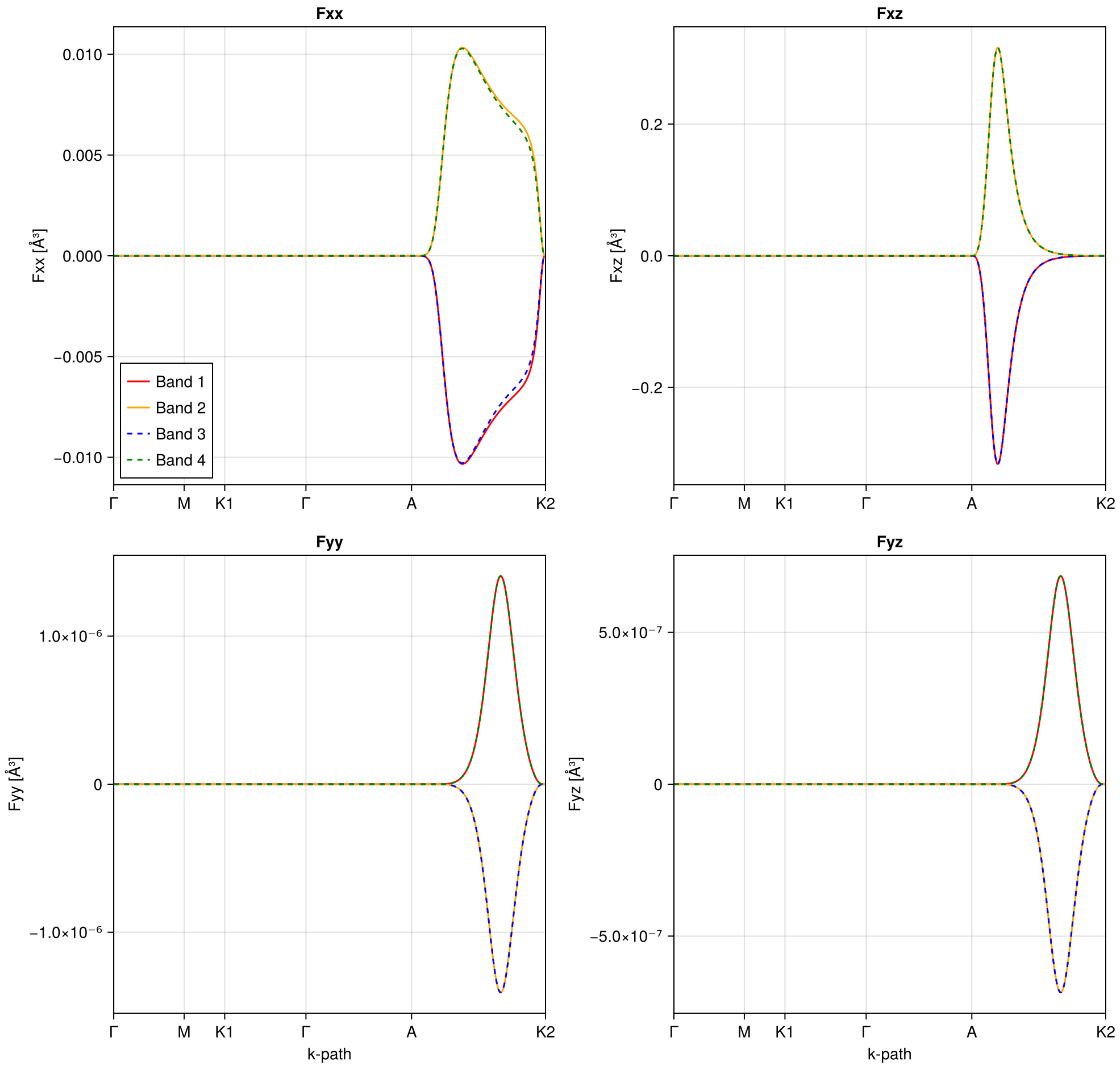

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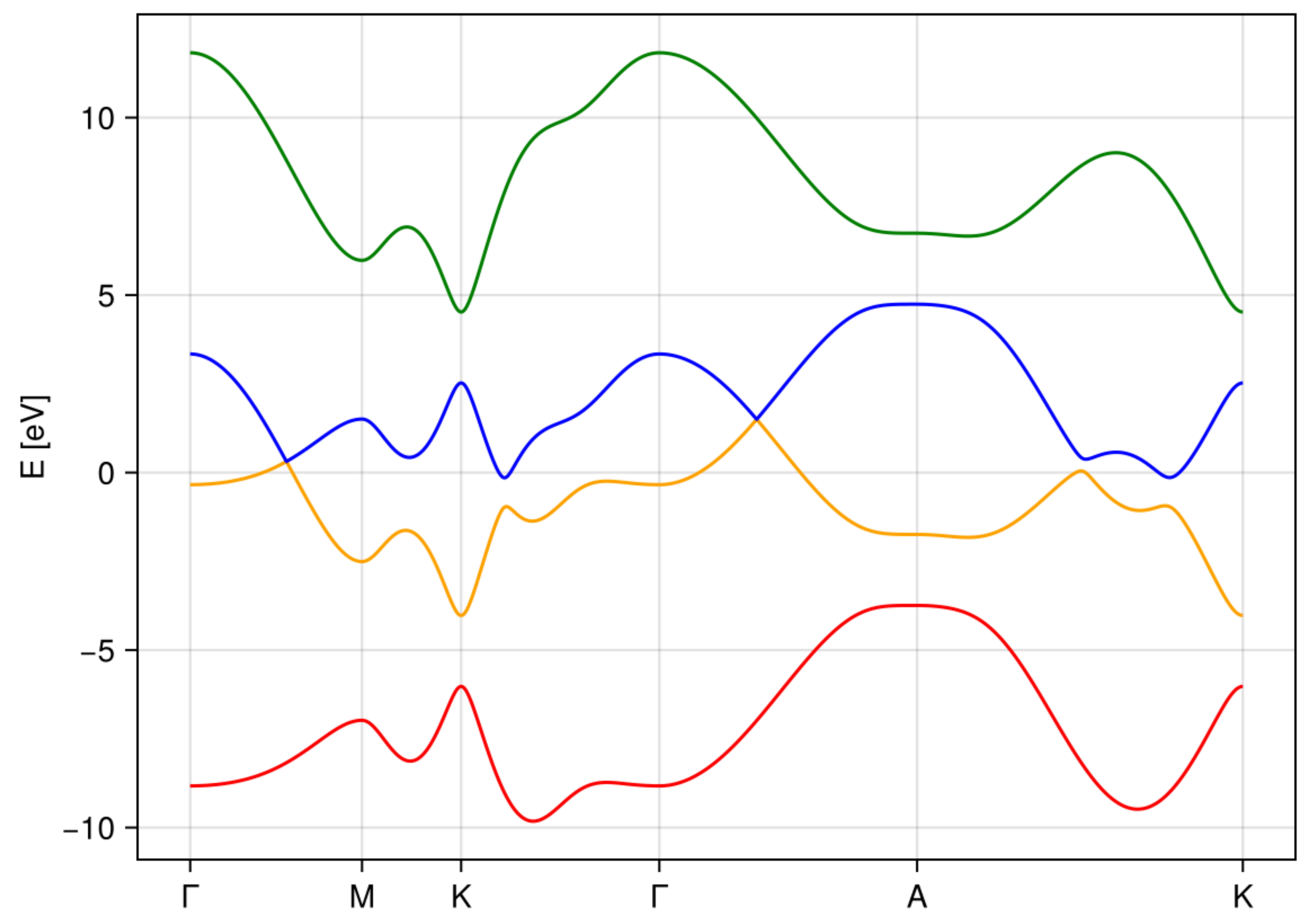


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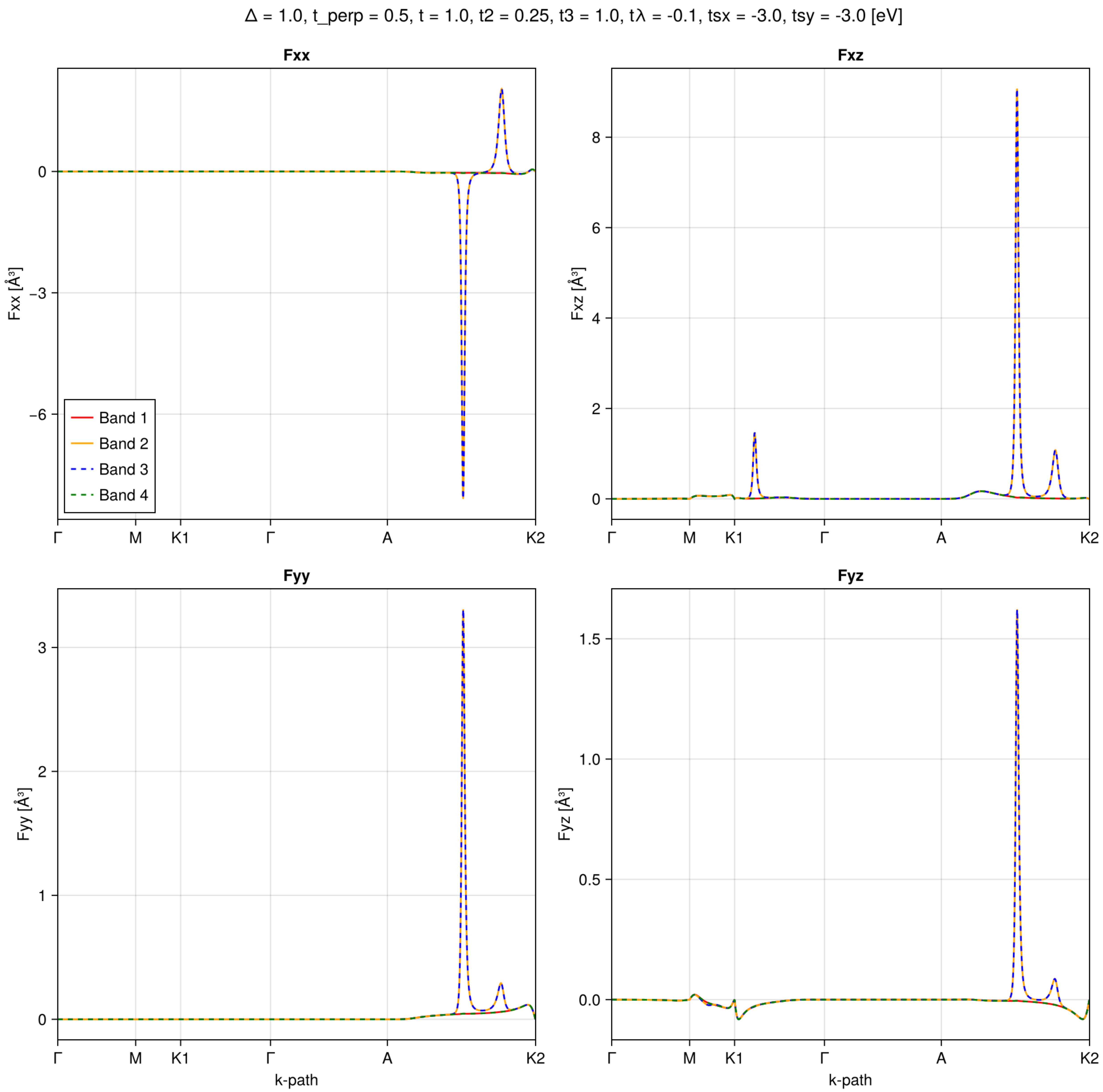


## OMM